

B.Tech. Degree IV Semester Examination May 2004

- IX. (a) With suitable diagrams explain the phenomenon of combustion in SI engines. (10)
- (b) Write short notes on the following:
- (i) Octane number (ii) Turbo charging
 - (iii) After burning (iv) HUCR
 - (v) Governing of IC engines (10)

OR

- X. (a) With neat sketch explain the working of Magneto-ignition system in SI engines. (10)
- (b) Sketch and explain the different combustion chambers of CI engines. (10)



ME 403 APPLIED THERMODYNAMICS

(2002 Admissions)

Time: 3 Hours

Maximum Marks: 100

- I. (a) State and explain First law of thermodynamics with illustrations. What are its limitations? (10)
- (b) Air at 101.3KPa, 20°C is taken into a gas turbine power plant at a velocity of 140 m/s through an opening of 0.15m² cross-sectional area. The air is compressed, heated, expanded through a turbine and exhausted at 0.18MPa, 150°C through an opening of 0.10m² cross-sectional area. The power output is 375KW. Calculate the net amount of heat added to the air in KJ/Kg. Assume that the air obeys the law $p v = 0.287(t + 273)$, where p is the pressure in KPa, v is the specific volume in m³/kg and ' t ' is the temperature in °C. Take $C_p = 1.005$ KJ/Kg K. (10)
- OR
- II. (a) State and prove 'Carnot's theorem'. (10)
- (b) A heat pump is used to heat a house in winter and then reversed to cool the house in summer. The interior temperature is to be maintained at 25°C during winter and summer. The heat transfer through walls and roof is estimated to be 2600KJ/hr per degree temperature difference between the inside and outside. Determine:
- (i) The minimum power required to drive the pump during winter if the outside temperature is 4°C.
 - (ii) The minimum power to drive the machine during summer if the outside temperature is 42°C. (10)
- III. (a) Explain the significance of Gibbs and Helmholtz functions. (5)
- (b) State Dalton's law and Amagat's law of mixture of perfect gases. (5)

(Turn over)

- (c) A perfect gas mixture consists of 1.4Kg of N_2 and 2.3Kg of CO_2 at a pressure of 2.8 bar and a temperature of $20^\circ C$. Find -
- The mole fraction of each constituent.
 - Equivalent molecular weight of mixture.
 - Equivalent gas constant of the mixture
 - Partial pressure and partial volume
 - Volume and density of the mixture. (10)

OR

- IV. (a) State and derive Gibbs-Dalton law for mixtures of ideal gases. (10)
- (b) The volumetric analysis of a gas is as follows:
 CO_2 40%, N_2 40% and O_2 20%. The pressure of gas is 1 bar and temperature is $15^\circ C$. Calculate the partial pressures of each constituent. (10)

- V (a) Sketch and explain the working of a Bomb calorimeter. (10)
- (b) The product of combustion of a hydrocarbon fuel of unknown composition have the following composition as measured on a dry basis.
 $CO_2 = 8.0\%$ $CO = 0.9\%$ $O_2 = 8.8\%$ $N_2 = 82.3\%$
 Calculate (i) the air fuel ratio (ii) the percent theoretical air on a mass basis. (10)

OR

- VI. (a) Explain the following:
- Adiabatic flame temperature
 - higher and lower heating values (10)
- (b) The lower heating value of propane at constant pressure and $25^\circ C$ is 20,440.09 KJ/Kg mole. Find the higher heating value at constant pressure and at constant volume. (10)

- VII. (a) Explain the working of Four-stroke petrol engine with reference to actual working cycle. (10)

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- (b) A four cylinder engine was tested at constant speed and throttle opening. The fuel supply was 0.089 Kg/min when all cylinders were firing. The spark-plugs were then successively short circuited without change of speed. The brake power measurements were as follows:

With all cylinders firing	- 15KW
Cylinder 1 cut out	- 9.04 KW
Cylinder 2 cut out	- 9.3 KW
Cylinder 3 cut out	- 9.32 KW
Cylinder 4 cut out	- 9.26KW

Determine the mechanical efficiency and indicated thermal efficiency of the engine if the calorific value of the fuel used is 41840 KJ/Kg. (10)

OR

VIII.

A 4-cylinder petrol engine has a bore of 57mm and a stroke of 90mm. The rated speed is 2800 rpm. The engine is tested at this speed against a brake which has a torque arm of 35.6cm. The net brake load is 155N and the fuel consumption is 6.74 litres/hr. The specific gravity of petrol used is 0.735 and it has a lower calorific value of 44200 KJ/Kg. A Morse test is carried out and the cylinders are cut out in order 1, 2, 3, 4 with corresponding brake loads 111, 106.5, 104.2 and 111N respectively. Calculate -

- The engine torque
- Brake mean effective pressure
- Brake thermal efficiency
- Specific fuel consumption
- Mechanical efficiency
- Indicated mean effective pressure (20)

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